

Convert the given file into docker compose :
FROM wordpress:latest
Set environment variables for database connection
ENV WORDPRESS_DB_HOST=db
ENV WORDPRESS_DB_USER=wpuser
ENV WORDPRESS_DB_PASSWORD=wppassword
ENV WORDPRESS_DB_NAME=wpdatabase
Expose port
EXPOSE 80

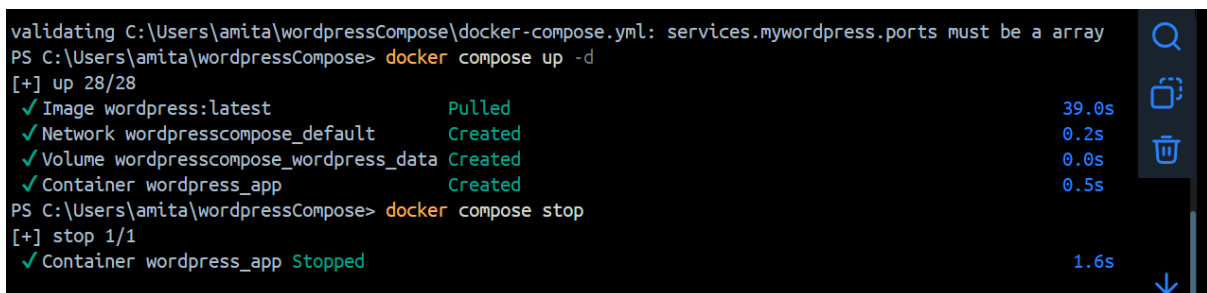
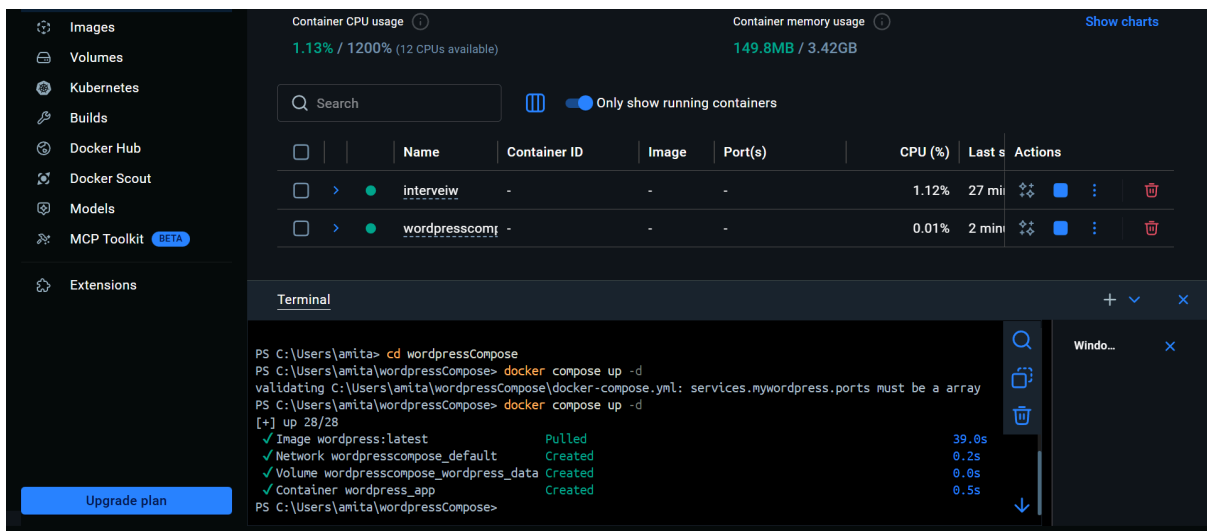
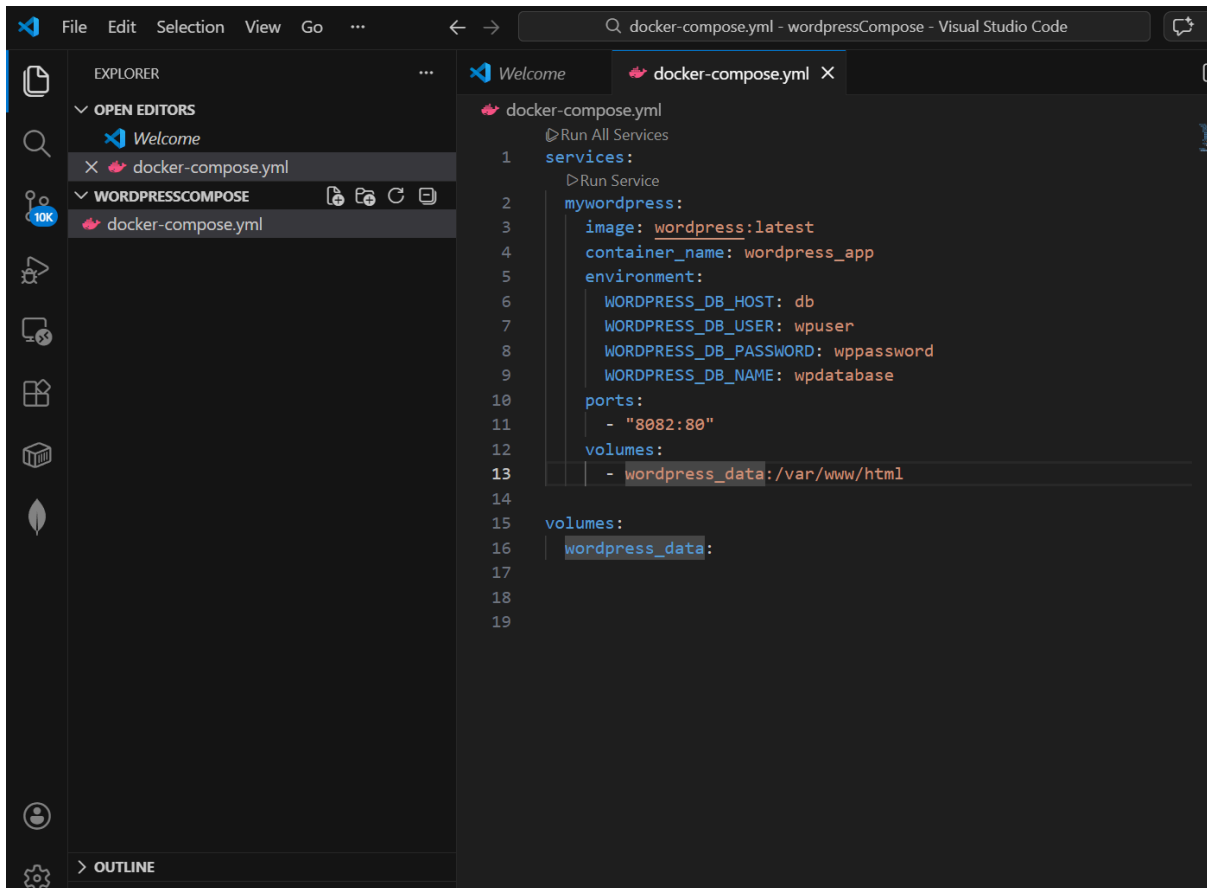
With docker run command :m
docker run -d \
--name wordpress_app \
-p 8082:80 \
-e WORDPRESS_DB_HOST=db \
-e WORDPRESS_DB_USER=wpuser \
-e WORDPRESS_DB_PASSWORD=wppassword \
-e WORDPRESS_DB_NAME=wpdatabase \
-v wordpress_data:/var/www/html \
mywordpress

```
PS C:\Users\amita> mkdir wordpressCompose

Directory: C:\Users\amita

Mode                LastWriteTime         Length Name
----                -
d-----          11-03-2026   12:20                wordpressCompose

PS C:\Users\amita> cd .\wordpressCompose\
PS C:\Users\amita\wordpressCompose> code .
PS C:\Users\amita\wordpressCompose> |
```



Practical No 4..>>>>>>

Create the following folder structure:

```
node-mongo-compose/  
|  
├── docker-compose.yml  
├── package.json  
├── server.js  
└── Dockerfile
```

file name : server.js

```
const express = require("express");  
const mongoose = require("mongoose");  
const app = express();  
const PORT = 3000;  
mongoose.connect("mongodb://mongo:27017/mydatabase", {  
  useNewUrlParser: true,  
  useUnifiedTopology: true  
})  
.then(() => console.log("MongoDB Connected"))  
.catch(err => console.log(err));  
app.get("/", (req, res) => {  
  res.send("Node.js + MongoDB running with Docker Compose");  
});  
app.listen(PORT, () => {  
  console.log(`Server running on port ${PORT}`);  
});
```

File name : package.json

```
{  
  "name": "node-mongo-app",  
  "version": "1.0.0",  
  "main": "server.js",  
  "dependencies": {  
    "express": "^4.18.2",  
    "mongoose": "^7.0.0"  
  }  
}
```

Dockerfile (for Node.js App)

```
FROM node:18  
WORKDIR /app  
COPY package.json .  
RUN npm install  
COPY . .  
EXPOSE 3000  
CMD ["node", "server.js"]
```

Docker Compose File

docker-compose.yml

version: '3.8'

services:

app:

build: .

container_name: node_app

ports:

- "3000:3000"

depends_on:

- mongo

restart: always

mongo:

image: mongo:6

container_name: mongo_db

ports:

- "27017:27017"

volumes:

- mongo_data:/data/db

restart: always

volumes:

mongo_data:

Steps to Run the Application

Step : Start containers

docker compose up --build

Step : Verify running containers

docker ps

```
🐳 docker-compose.yml
1  version: '3.8'
2
3  Run All Services
4  services:
5    Run Service
6    app:
7      build: .
8      container_name: node_app
9      ports:
10       - "3000:3000"
11      depends_on:
12       - mongo
13      restart: always
14
15    Run Service
16    mongo:
17      image: mongo:6
18      container_name: mongo_db
19      ports:
20       - "27017:27017"
21      volumes:
22       - mongo_data:/data/db
23      restart: always
24
25 volumes:
26   mongo_data:
```

```
🐳 Dockerfile > ...
1  FROM node:18
2  WORKDIR /app
3  COPY package.json .
4  RUN npm install
5  COPY . .
6  EXPOSE 3000
7  CMD ["node", "server.js"]
```

```
{ } package.json > ...
1  {
2    "name": "node-mongo-app",
3    "version": "1.0.0",
4    "main": "server.js",
5    "dependencies": {
6      "express": "^4.18.2",
7      "mongoose": "^7.0.0"
8    }
9  }
```

```
JS server.js > ...
1  const express = require("express");
2  const mongoose = require("mongoose");
3  const app = express();
4  const PORT = 3000;
5  mongoose.connect("mongodb://mongo:27017/mydatabase", {
6    useNewUrlParser: true,
7    useUnifiedTopology: true
8  })
9  .then(() => console.log("MongoDB Connected"))
10 .catch(err => console.log(err));
11 app.get("/", (req, res) => {
12   res.send("Node.js + MongoDB running with Docker Compose");
13 });
14 app.listen(PORT, () => {
15   console.log(`Server running on port ${PORT}`);
16 });
```

```
PS C:\Users\amita\My Project\docker\node-mongo-compose> docker compose up --build
time="2026-03-12T12:47:37+05:30" level=warning msg="C:\\Users\\amita\\My Project\\docker\\node-mongo-compo
se\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid
potential confusion"
#1 [internal] load local bake definitions
#1 reading from stdin 575B done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 151B 0.0s done
#2 DONE 0.0s
```

RAM 3.26 GB CPU 0.25% Disk: 62.53 GB used (limit 1006.85 GB)

```
Terminal
node_app | heartbeatFrequencyMS: 10000,
node_app | localThresholdMS: 15,
node_app | setName: null,
node_app | maxElectionId: null,
node_app | maxSetVersion: null,
node_app | commonWireVersion: 0,
node_app | logicalSessionTimeoutMinutes: null
node_app | },
node_app | code: undefined
node_app | }
mongo_db | {"t":{"$date":"2026-03-12T07:18:42.708+00:00"},"s":"I", "c":"WTCHKPT", "id":22430, "ctx":"
Checkpoint", "msg":"WiredTiger message", "attr":{"message":{"ts_sec":1773299922, "ts_usec":706654, "thread":
"1:0x7ddba87df640", "session_name":"WT_SESSION.checkpoint", "category":"WT_VERB_CHECKPOINT_PROGRESS", "catego
ry_id":6, "verbose_level":"DEBUG", "verbose_level_id":1, "msg":"saving checkpoint snapshot min: 33, snapshot
max: 33 snapshot count: 0, oldest timestamp: (0, 0) , meta checkpoint timestamp: (0, 0) base write gen: 1"
}}}
```

RAM 3.25 GB CPU 0.92% Disk: 62.53 GB used (limit 1006.85 GB) v4.64.0



Node.js + MongoDB running with Docker Compose

Question 1: The following configuration runs as Apache HTTP server.
version: "3"

Services:

Web:

Image:httpd

Modify the file to:

1. Map port 8081 on host to port 80 in container
2. Mount a local folder ./website to /usr/local/apache2/htdocs
3. Set container name as apache_server

Question 2: Create a docker-compose.yml file to deploy the following application:

Requirements

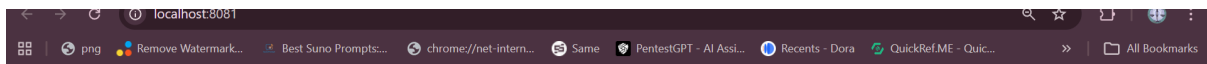
1. Service 1: Nginx Web server

Image: nginx

Port Mapping: 8085:80

2. Service 2: Redis Database
Image:redis
3. Both services must:
Run on a custom network called appnetwork
4. The nginx container name should be nginx_server

```
tainer to be able to reuse that name.
PS C:\Users\amita\My Project\docker\ngrnix-copose> ^C
PS C:\Users\amita\My Project\docker\ngrnix-copose> docker rm -f apache_server
apache_server
PS C:\Users\amita\My Project\docker\ngrnix-copose> docker-compose up -d
time="2026-03-13T12:45:10+05:30" level=warning msg="C:\\Users\\amita\\My Project\\docker\\ngrnix-copose\\do
cker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid poten
tial confusion"
[+] up 1/1
✓ Container apache_server Started 0.9s
PS C:\Users\amita\My Project\docker\ngrnix-copose> 
```



Index of/



```
docker-compose.yml X  Release Notes: 1.111.0
C: > Users > amita > My Project > docker > nginx-copose > docker-compose.yml
1  version: "3"
   Run All Services
2  services:
   Run Service
3    nginx:
4      image: nginx
5      container_name: nginx_server
6      ports:
7        - "8085:80"
8      networks:
9        - appnetwork
10
11   Run Service
12   redis:
13     image: redis
14     networks:
15       - appnetwork
16
17 networks:
18   appnetwork:
19     driver: bridge
```

```
230e39d8034f  none  null  local
a52f72bceca3  wordpresscompose_default  bridge  local
PS C:\Users\amita\My Project\docker\nnginx-copose> docker-compose up -d
time="2026-03-13T12:49:06+05:30" level=warning msg="C:\\Users\\amita\\My Project\\docker\\nginx-copose\\do
cker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid poten
tial confusion"
[+] up 8/9
✓ Image redis  Pulled 11.2s
- Network nginx-copose_appnetwork Creating 0.1s
[+] up 11/113-13T12:49:17+05:30" level=warning msg="Found orphan containers ([apache_server]) for this pro
✓ Image redis  Pulled 11.2s
RAM 1.57 GB CPU 0.25% Disk: 62.59 GB used (limit 1006.85 GB)
```

```
>>
>> # Inspect which containers are on appnetwork
>> docker network inspect appnetwork
CONTAINER ID  IMAGE  COMMAND  NAMES  CREATED  STATUS
PORTS
fd3a81c80507  nginx  "/docker-entrypoint..."  About a minute ago  Up About a minute
0.0.0.0:8085->80/tcp, [::]:8085->80/tcp  nginx_server
ca7123deda30  redis  "docker-entrypoint.s..."  About a minute ago  Up About a minute
6379/tcp  nginx-copose-redis-1
d84772bf02f0  httpd  "httpd-foreground"  5 minutes ago  Up 5 minutes
0.0.0.0:8081->80/tcp, [::]:8081->80/tcp  apache_server
RAM 1.32 GB CPU 0.67% Disk: 62.59 GB used (limit 1006.85 GB)
```

```
PS C:\Users\amita\My Project\docker\nginx-copose> docker ps
CONTAINER ID   IMAGE                                COMMAND                                  CREATED        STATUS
PORTS         NAMES
fd3a81c80507   nginx                              "/docker-entrypoint...."              About a minute Up About a minute
0.0.0.0:8085->80/tcp, [::]:8085->80/tcp  nginx_server
ca7123deda30   redis                              "docker-entrypoint.s..."            About a minute Up About a minute
6379/tcp      nginx-copose-redis-1
d84772bf02f0   httpd                              "httpd-foreground"                   6 minutes ago  Up 6 minutes
0.0.0.0:8081->80/tcp, [::]:8081->80/tcp  apache_server
740cb784f755   node-mongo-compose-app            "docker-entrypoint.s..."            24 hours ago   Up 28 minutes

RAM 1.32 GB   CPU 1.25%   Disk: 62.59 GB used (limit 1006.85 GB)
```

```
PS C:\Users\amita\My Project\docker\nginx-copose> docker network ls
NETWORK ID     NAME                                DRIVER  SCOPE
693c18ab21ab   amiai_default                      bridge  local
fadcd45eefa7   bridge                            bridge  local
e246c576fc12   docker-compose_default            bridge  local
02c724f7efcf   gabizap-network                   bridge  local
c5bbd7727bdd   host                              host    local
49921a945773   interveiw_default                 bridge  local
ecad051b27d9   nginx-copose_appnetwork           bridge  local
26b11deeb9fb   nginx-copose_default              bridge  local
a199cb9046e4   node-mongo-compose_default        bridge  local

RAM 1.32 GB   CPU 0.25%   Disk: 62.59 GB used (limit 1006.85 GB)
```

